Course Code	**ESTIMATION, SPECIFICATIONS, COSTING AND VALUATION**	L	T	P	C
23A01507		0	1	2	2

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the various methods and types of estimates usedin civil engineering projects.
2. **Develop** detailed estimates for single and multi-storey buildings using standard estimation methods.
3. **Analyze** rate analysis, abstract estimation, and bill preparation as per standard procedures.
4. **Prepare** detailed specifications and tender documents for construction works.
5. **Evaluate** the valuation, cost escalation, and value analysis of buildings.

Course Outcomes:

Upon successful completion of this course, students will be able to:

1. **Apply** estimation techniques to prepare detailed estimates for various construction projects.
2. **Develop** abstract estimates and rate analysis for different civil engineering works.
3. **Analyze** the preparation of measurement books and bill preparation as per AP State Government procedures.
4. **Create** detailed specifications and tender documents for construction projects.
5. **Assess** building valuation, cost escalation, and value analysis techniques.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	2	1	-	-	-	-	-	1	-	2	3	2
CO -2	3	3	3	2	2	-	-	-	-	-	-	1	3	3
CO -3	3	3	3	2	2	-	2	1	-	-	-	1	3	3
CO -4	3	3	3	2	2	-	2	1	-	-	-	1	3	3
CO -5	2	2	2	1	3	-	3	1	-	1	1	2	2	2

List of Experiments

1. Activity Based On Learning Methods and Types of Estimates
2. Preparation of Detailed Estimate for A Single-Storied Residential Building Using Wall to Wall Method
3. Preparation of Detailed Estimate for A Single Storied Residential Building Using Centre Line Method for Earthwork, Foundations, Super Structure, Fittings Including Sanitary and Electrical Fittings &Paintings.
4. Preparation of Detailed Estimate for A Two Storied Residential Building Using Centre Line Method for Earthwork, Foundations, Super Structure, Fittings Including Sanitary and Electrical Fittings &Paintings.

5. Activity Based Learning of Estimate Data and Rate Analysis
6. Preparation of Abstract Estimate for The Detailed Estimate in Exercise No.3
7. Preparation of Abstract Estimate for The Detailed Estimate in Exercise No.4
8. Writing of Measurement Book and Bill Preparation as Per AP State Govt Procedure for Detailed Estimate in No. 3 and Abstract Estimate of No. 6
9. Writing of Detailed Specifications for Various Items of Estimate and Preparing A Model Tender Document for The Work Listed in No. 3 and 6
10. Activity Based Learning for Valuation of Buildings, Cost Escalation Procedures and Value Analysis for Any One Work

TEXT BOOKS:

1. B.N. Dutta - Estimating and Costing in Civil Engineering, CBS Publishers & Distributors, 28th Revised Edition (2020).
2. Rangwala - Estimating, Costing and Valuation, Charotar Publishing House, 2023.
3. D.D. Kohli& R.C. Kohli - A Textbook of Estimating and Costing (Civil), S. Chand Publishing, 2011.

REFRENCE BOOKS:

1. M. Chakraborti - Estimating, Costing, Specification &Valuationin Civil Engineering, 29th Edition (2021).
2. Gurcharan Singh - Estimating, Costing and Valuation, Standard Publishers, 2018.
3. V.N. Vazirani& S.P. Chandola - Civil Engineering Estimating & Costing, Khanna Publishers, 4th Edition (2001).

Online Learning Resources:

https://onlinecourses.swayam2.ac.in/nou20_cs11/preview
https://www.coursera.org/learn/construction-cost-estimating